Numerical Analysis (2627-MA385): **Class Test**

When:	15:00 Thursday, 29 October 2026 (CSB-1005 and AC201).
Duration:	40 minutes.
Instructions:	Answer all questions in the answer book provided
Don't forget:	to write your name and ID number on the answer book.
Rules:	No notes, books or log tables allowed. Only basic calculators permitted. Don't sit beside anybody else.

.....

Q1. (a) State the *Secant Method* for solving the nonlinear equation $f(x) = 0$.

(b) Give a derivation (or careful justification) for this method.

(c) We wish to use the Secant Method to estimate $\tau = \sqrt[3]{4}$ numerically by solving $f(x) = 0$ in $[a, b]$ for some suitably chosen f, a and b. Suggest suitable choices of f, a, and b for this problem, and show that f has a zero in $[a, b]$.

(d) Take $x_0 = a$ and $x_1 = b$. use 3 steps of the Secant Method method to estimate $\sqrt[3]{4}$. (That is, compute the estimates x_2, x_3 and x_4).

.....

Q2. (a) State *Newton's method* for solving the nonlinear equation $f(x) = 0$ *for some* $x \in [a, b]$.

(b) Give a derivation of Newton's Method, and the Newton Error formula, using a truncated Taylor Series.

(c) Generally, what do you expect the Order of Convergence to be for Newton's Method, and how is that justified by the Newton Error Formula? *Note: you don't have to prove convergence, just give a justification.*

(d) Suppose we take $f(x) = x^2$, and apply Newton's method to solve $f(x) = 0$ with $x_0 = 1$. What order of convergence does one observe in practice? Does this agree with your answer from Part (c)? If not, offer an explanation as to why.

.....

Q3. (a) State *Euler's method* for solving initial value problems:

$$y(t_0) = y_0 \qquad \text{and } y'(t) = f(t, y) \text{ for } t > t_0.$$

(b) Show how Euler's method can be motivated by using a truncated Taylor Series.

(c) Give an example of an Initial Value problem for which Euler's method would give the exact solution (i.e., with no error). Justify your answer with the Taylor Series you used to motivate Euler's method.